

Supplement for RDD results

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Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
- – robust rd not sig, donut RDD sig for 2 “larger holed donuts”, ordinary OLS is sig with 0.08 threshold,

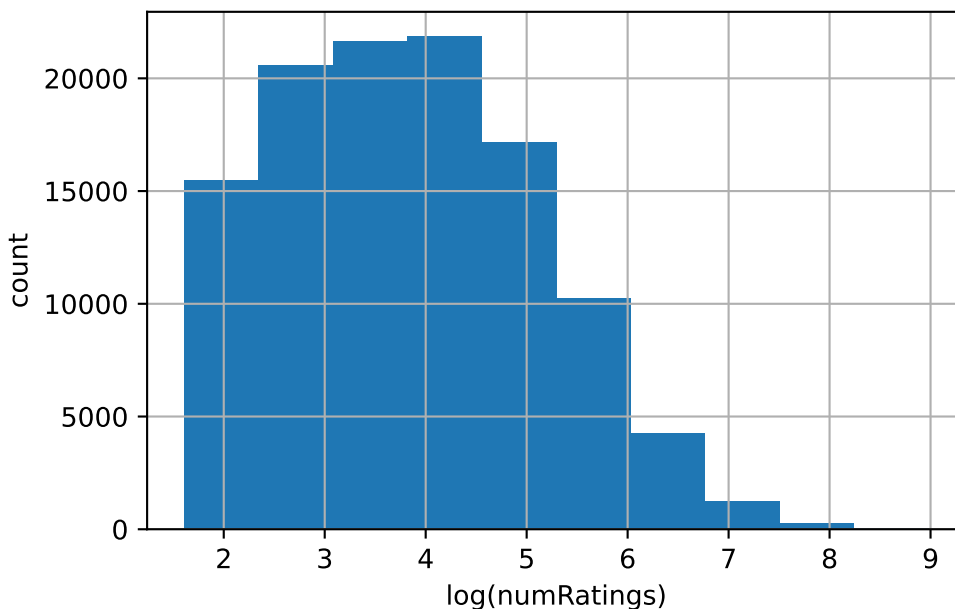
2 Setup & Data

```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 203392 notes written by our 3 choosen algorithms
Now we have 147890 notes classified as MISLEADING
Now we have 112758 notes which have enough ratings
----- No filter for alignment
FINAL total 112758 Notes
of which author writes again = 77111
```

NB current (final), not Max, score

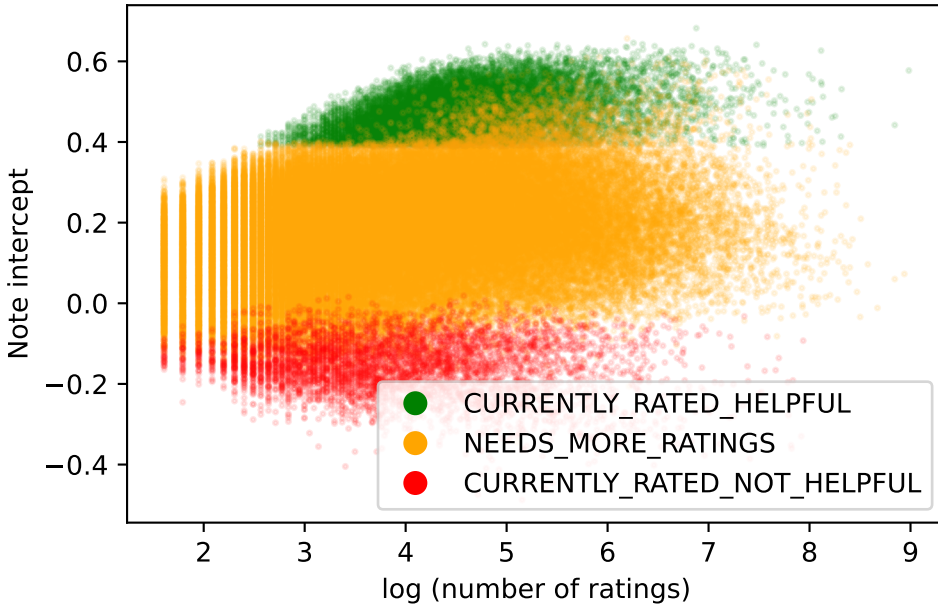
3 Diagnostic Plots

Most notes have a small number of rating, a small number have very many ratings This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting log(numRatings) vs NoteIntercept, with colours according to Note final status (finalRatingStatus)



Sadly, though, our data record intercept calculated from the full set of ratings, this is not the contemporaneous intercept score which determines moment-by-moment publication. It approximates it, but with contamination. Notes may be published then unpublished as they receive more ratings and their rating degrades, or there may be additional blocks on publication (e.g. requirements on total number or type of ratings, or requirements on intercept stability or a lag before publication). The CN system is not entirely clear about all these details.

These contamination can be seen by plotting indicators of ever having been published against our (final) intercept.

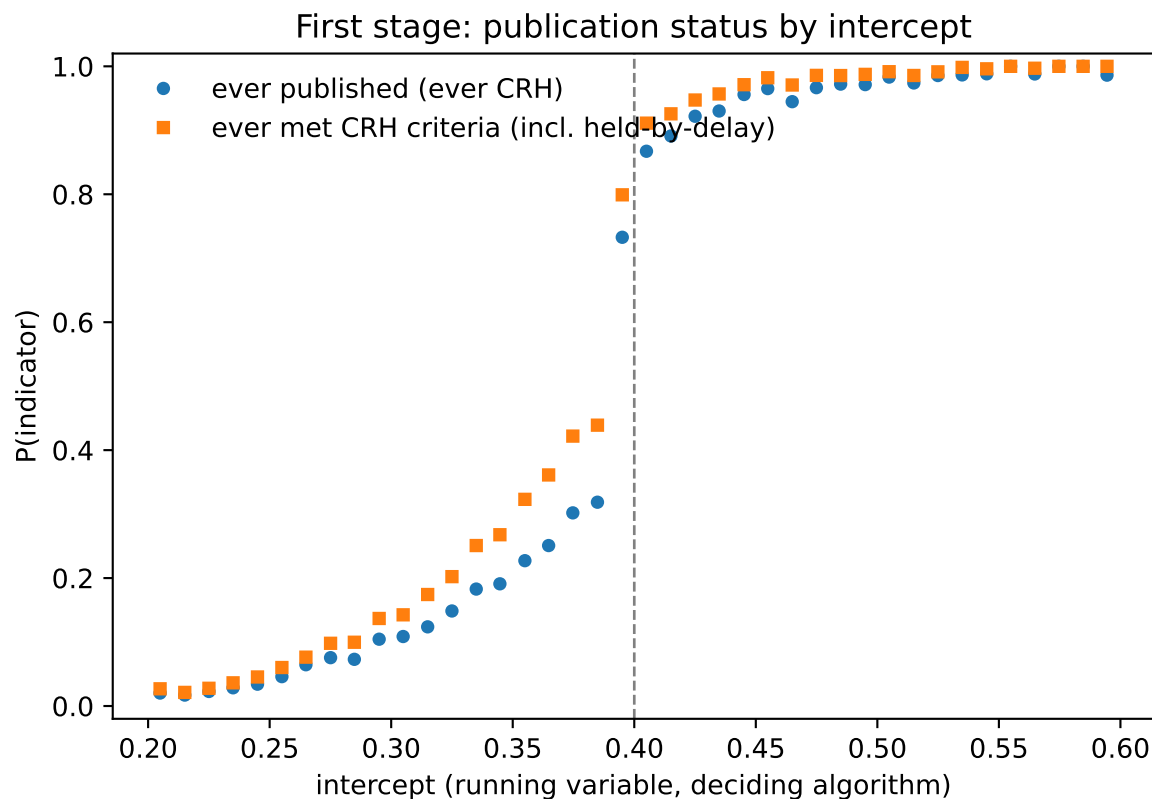
```
[0.35,0.40): n= 4547 P(ever_pub)=0.3851 P(ever_crit)=0.4841
[0.38,0.40): n= 1801 P(ever_pub)=0.5819 P(ever_crit)=0.6680
[0.39,0.40): n= 1145 P(ever_pub)=0.7328 P(ever_crit)=0.7991
[0.40,0.42): n= 2030 P(ever_pub)=0.8788 P(ever_crit)=0.9182
```

Below-cutoff P(if_written_again) by ever_published:

	mean	size
ever_published		
False	0.674867	91704
True	0.740338	4321

Below-cutoff P(if_written_again) by ever_met_crh_criteria:

	mean	size
ever_met_crh_criteria		
False	0.676021	90438
True	0.706819	5587



This shows that our secondary indicator of publication status shows a discontinuity near the cutoff, but not a sharp one

The high proportion of ‘ever published’ just below the cutoff suggests many notes got published and then slipped back below the threshold

We compromise with a fuzzy RDD

4 Trad OLS RDD

Just using OLS and an arbitrary window

bandwidth = 0.08

Number of treated units: 8198

Number of control units: 8829

Estimated Treatment Effect at the Cutoff: 0.0192

I think we can interpret this as a % change the chance of authoring again if published

P-value for treatment: 0.1927

T-statistic for treatment: 1.3027

5 RDD using rdrobust

This has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell, Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

running variable (x) length 112758

binary outcome (y) length 112758

cutoff (c) is 0.4

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	110954
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	96025	14929
Number of Unique Obs.	95875	14909
Number of Effective Obs.	4102	4723
Bandwidth Estimation	0.046	0.046
Bandwidth Bias	0.086	0.086
rho (h/b)	0.532	0.532

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.023	0.021	-1.111	2.664e-01	[-0.064, 0.018]
Robust	-	-	-1.373	1.696e-01	[-0.078, 0.014]

MSE-optimal $h = 0.0459$

Bias-corrected point estimate: -0.0321 95% CI $[-0.0779, 0.0137]$

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

$h = 0.0229$ (0.5x)	$\tau = -0.0394$	95% CI $[-0.1095, +0.0207]$	$N_L/N_R = 1993/2323$
$h = 0.0459$ (1.0x)	$\tau = -0.0233$	95% CI $[-0.0779, +0.0137]$	$N_L/N_R = 4102/4723$
$h = 0.0917$ (2.0x)	$\tau = +0.0053$	95% CI $[-0.0334, +0.0330]$	$N_L/N_R = 10844/9320$

Mephi suggested text :

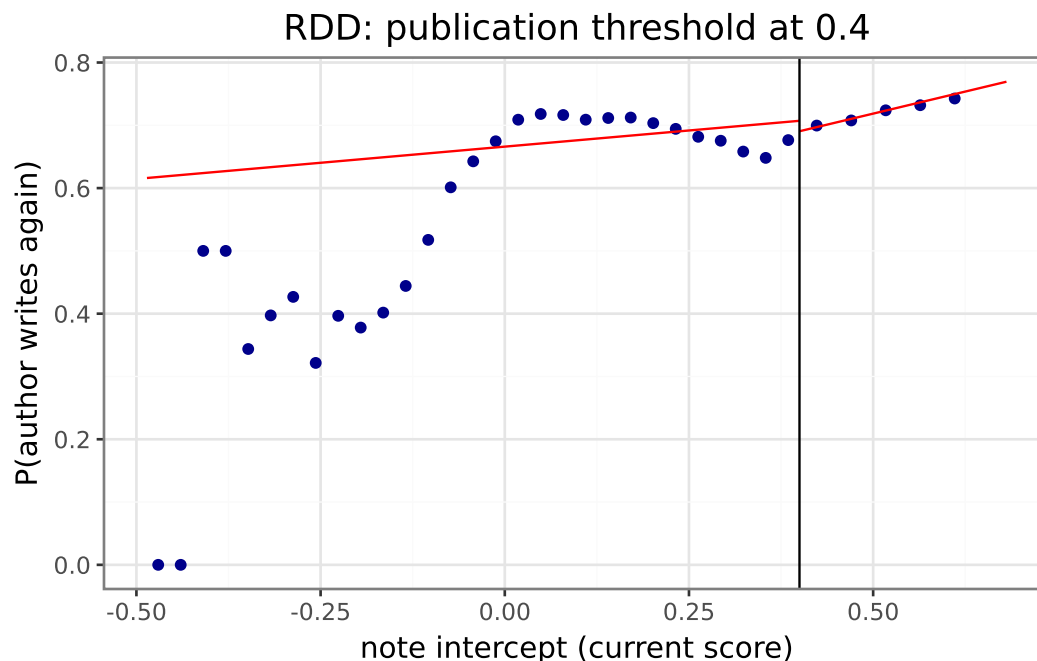
At the MSE-optimal bandwidth ($h = 0.046$) we estimate $\hat{\tau} = -2.3\text{pp}$ in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI $[-7.8, +1.4]$, $p = 0.17$).

6 Converting rate to baseline outcome rate

Mephi suggested text: Untreated authors at the cutoff re-author at 66.5%; treated authors at the cutoff re-author at ~64.2%. Treatment effect = -2.3pp , a -3.5% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with global linear fit (this is the default plot from `rdplot`)



`['rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdplot_paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (34 bins)`

RD plot at $c = 0.4$ ($p=1$, triangular, $N=112758$)

```

bins used      : 29 left / 6 right
J_IMSE        : [29.0, 6.0]
J_MV          : [883.0, 887.0]
scale         : [1, 1]
bin width      : avg [0.03054120705045503, 0.0468238353729248], median [0.030541207050455044, 0

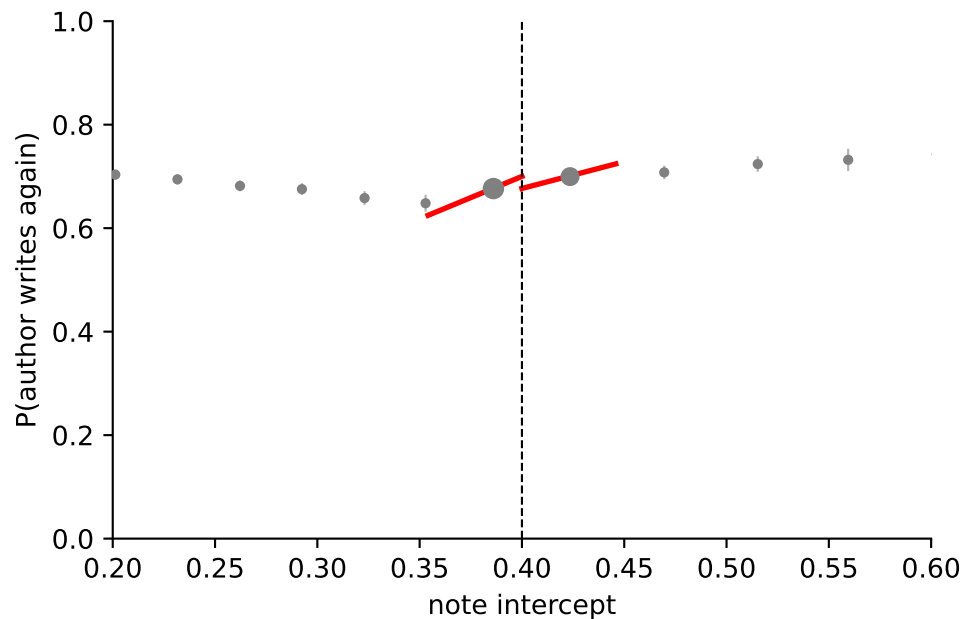
```

coefficients:

	Left	Right
0	0.707181	0.690543
1	0.102853	0.280793

wrote paper-outputs/RDDrobust_scatter_meta.json, paper-outputs/RDDrobust_scatter_coef.csv

Now with the local-linear fits (over the rdrobust defined window)



consistency check: gap at cutoff should be conventional tau
plot gap at $c = -0.0233$ vs conventional tau = -0.0233

8 Robustness check - density test

Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

Inspecting density with a window of ± 0.02

```

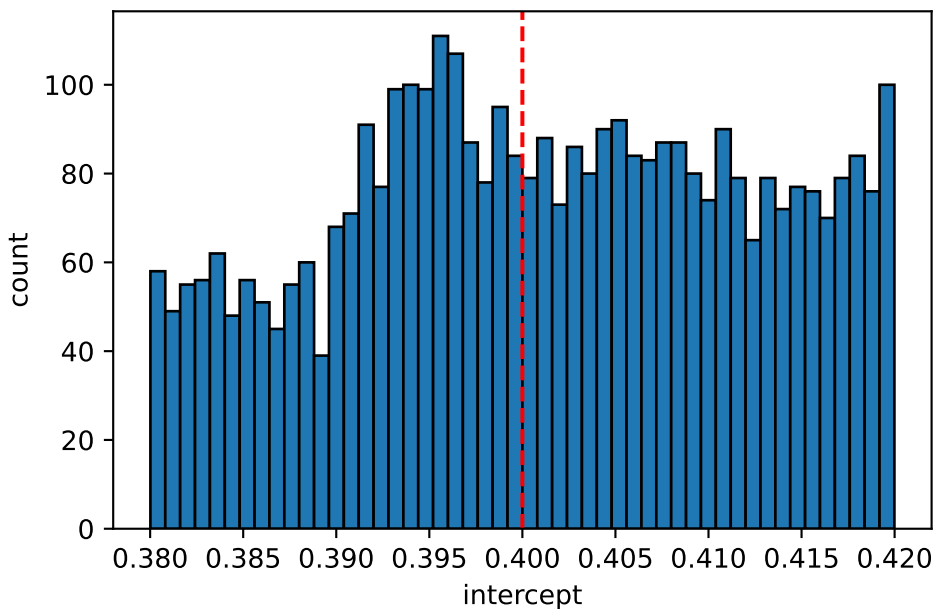
t_asy      NaN
t_jk      -4.215801
p_asy      NaN
p_jk       0.000025
dtype: float64

```

```

left      1.316348
right     0.994493
diff      -0.321855
dtype: float64
left      0.041508
right     0.040941
dtype: float64
full      110954
left      96025
right     14929
eff_left   3633
eff_right  4240
dtype: int64
leftN      [1801.0, 1957.0, 2136.0, 2343.0, 2538.0, 2731....
rightN     [2030.0, 2274.0, 2494.0, 2741.0, 2997.0, 3248....
leftWindow  [0.02, 0.022389783198856886, 0.024779566397713...
rightWindow [0.02, 0.02232674639281102, 0.0246534927856220...
pval       [0.00022900639310549447, 1.1723340901379433e-0...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino']

```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Excluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	4102	4723	0.046	-0.0233	0.0210	0.1696	-0.0779	0.0137

±0.005	4736	5399	0.057	0.0056	0.0242	0.9339	-0.0574	0.0528
±0.010	8310	7519	0.084	0.0470	0.0221	0.0984	-0.0085	0.0996
±0.020	12875	8913	0.111	0.0493	0.0210	0.0850	-0.0067	0.1043

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

running variable (x) length 112758

binary outcome (y) length 112758

cutoff (c) is 0.3

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	110954
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	83588	27366
Number of Unique Obs.	83460	27324
Number of Effective Obs.	13567	8704
Bandwidth Estimation	0.058	0.058
Bandwidth Bias	0.098	0.098
rho (h/b)	0.588	0.588

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.011	0.014	0.816	4.146e-01	[-0.016, 0.038]
Robust	-	-	0.96	3.370e-01	[-0.016, 0.046]

MSE-optimal h = 0.0576

Bias-corrected point estimate: 0.0152 95% CI [-0.0158, 0.0462]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0288 (0.5x)	tau = +0.0246	95% CI [-0.0135, +0.0722]	N_L/N_R = 6371/5023
h = 0.0576 (1.0x)	tau = +0.0111	95% CI [-0.0158, +0.0462]	N_L/N_R = 13567/8704
h = 0.1153 (2.0x)	tau = +0.0000	95% CI [-0.0152, +0.0292]	N_L/N_R = 30038/13984

running variable (x) length 112758

binary outcome (y) length 112758
cutoff (c) is 0.35
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 110954
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	91478	19476
Number of Unique Obs.	91333	19451
Number of Effective Obs.	21263	10492
Bandwidth Estimation	0.107	0.107
Bandwidth Bias	0.164	0.164
rho (h/b)	0.653	0.653

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.003	0.013	-0.208	8.353e-01	[-0.029, 0.023]
Robust	-	-	-0.411	6.809e-01	[-0.037, 0.024]

MSE-optimal h = 0.1069

Bias-corrected point estimate: -0.0065 95% CI [-0.0372, 0.0243]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0535 (0.5x)	tau = +0.0026	95% CI [-0.0372, +0.0506]	N_L/N_R = 8631/4894
h = 0.1069 (1.0x)	tau = -0.0027	95% CI [-0.0372, +0.0243]	N_L/N_R = 21263/10492
h = 0.2138 (2.0x)	tau = +0.0052	95% CI [-0.0195, +0.0266]	N_L/N_R = 51948/18548

running variable (x) length 14929

binary outcome (y) length 14929

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 14929
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	5187	9742
Number of Unique Obs.	5180	9729
Number of Effective Obs.	1487	1466
Bandwidth Estimation	0.014	0.014
Bandwidth Bias	0.023	0.023
rho (h/b)	0.627	0.627

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.062	0.037	1.705	8.826e-02	[-0.009, 0.134]
Robust	-	-	1.659	9.711e-02	[-0.013, 0.156]

MSE-optimal h = 0.0142

Bias-corrected point estimate: 0.0716 95% CI [-0.0130, 0.1561]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0071 (0.5x) tau = +0.0878 95% CI [-0.0244, +0.2151] N_L/N_R = 754/778
h = 0.0142 (1.0x) tau = +0.0623 95% CI [-0.0130, +0.1561] N_L/N_R = 1487/1466
h = 0.0284 (2.0x) tau = +0.0438 95% CI [-0.0061, +0.1132] N_L/N_R = 2990/2851

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 14929
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	5187	9742
Number of Unique Obs.	5180	9729
Number of Effective Obs.	1487	1466
Bandwidth Estimation	0.014	0.014
Bandwidth Bias	0.023	0.023
rho (h/b)	0.627	0.627

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.062	0.037	1.705	8.826e-02	[-0.009, 0.134]
Robust	-	-	1.659	9.711e-02	[-0.013, 0.156]

All these are non-significant

11 Sensitivity check - data for different time windows

PERIOD 1, from 2025-04-08 to 2026-06-01

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 87791 first time note authors 2025-04-08 to 2026-06-01

Now we have 68536 notes written by our 3 choosen algorithms

Now we have 52423 notes classified as MISLEADING

Now we have 34862 notes which have enough ratings

----- No filter for alignment

FINAL total 34862 Notes

of which author writes again = 19514

NB current (final), not Max, score

running variable (x) length 34862

binary outcome (y) length 34862

cutoff (c) is 0.4

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	34856
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	29401	5455
Number of Unique Obs.	29391	5448
Number of Effective Obs.	2460	2627
Bandwidth Estimation	0.064	0.064
Bandwidth Bias	0.099	0.099
rho (h/b)	0.642	0.642

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.001	0.03	-0.036	9.712e-01	[-0.06, 0.058]
Robust	-	-	-0.253	8.000e-01	[-0.079, 0.061]

MSE-optimal h = 0.0636

Bias-corrected point estimate: -0.0090 95% CI [-0.0787, 0.0607]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):
 h = 0.0318 (0.5x) tau = -0.0148 95% CI [-0.1174, +0.0793] N_L/N_R = 1050/1320
 h = 0.0636 (1.0x) tau = -0.0011 95% CI [-0.0787, +0.0607] N_L/N_R = 2460/2627
 h = 0.1271 (2.0x) tau = +0.0228 95% CI [-0.0381, +0.0637] N_L/N_R = 6786/4626

PERIOD 1, from 2024-01-01 to 2025-04-07

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 172844 first time note authors 2024-01-01 to 2025-04-07

Now we have 134572 notes written by our 3 choosen algorithms

Now we have 95257 notes classified as MISLEADING

Now we have 77741 notes which have enough ratings

----- No filter for alignment

FINAL total 77741 Notes

of which author writes again = 57520

NB current (final), not Max, score

running variable (x) length 77741

binary outcome (y) length 77741

cutoff (c) is 0.4

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 75944
 Polynomial Order Est. (p): 1
 Polynomial Order Bias (q): 2
 Kernel: Triangular
 Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	66486	9458
Number of Unique Obs.	66393	9456
Number of Effective Obs.	3370	3667
Bandwidth Estimation	0.059	0.059
Bandwidth Bias	0.103	0.103
rho (h/b)	0.568	0.568

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.006	0.022	-0.262	7.935e-01	[-0.049, 0.038]
Robust	-	-	-0.577	5.641e-01	[-0.063, 0.035]

MSE-optimal h = 0.0586

Bias-corrected point estimate: -0.0144 95% CI [-0.0634, 0.0345]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0293 (0.5x)	tau = -0.0233	95% CI [-0.0979, +0.0389]	N_L/N_R = 1542/1770
h = 0.0586 (1.0x)	tau = -0.0058	95% CI [-0.0634, +0.0345]	N_L/N_R = 3370/3667
h = 0.1171 (2.0x)	tau = +0.0196	95% CI [-0.0214, +0.0503]	N_L/N_R = 10056/7030